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***Additional sex combs-like 1* belongs to the enhancer of trithorax and Polycomb Group and genetically interacts with *Cbx2* in mice**

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Abstract

The *Additional sex combs* (*Asx*) gene of *Drosophila* behaves genetically as an enhancer of trithorax and Polycomb (ETP) in displaying bidirectional homeotic phenotypes, suggesting that is required for maintenance of both activation and silencing of *Hox* genes. There are 3 murine homologs of *Asx* called *Additional sex combs-like1*, 2, and 3. *Asx1* is required for normal adult hematopoiesis; however its embryonic function is unknown. We used a targeted mouse mutant line *Asx1*^{tm1Bc} to determine if *Asx1* is required to silence and activate *Hox* genes in mice during axial patterning. The mutant embryos exhibit simultaneous anterior and posterior transformations of the axial skeleton, consistent with a role for *Asx1* in activation and silencing of *Hox* genes. Transformations of the axial skeleton are enhanced in compound mutant embryos for the Polycomb group gene *M33/Cbx2*. *Hox a4*, *a7*, and *c8* are derepressed in *Asx1*^{tm1Bc} mutants in the antero-posterior axis, but *Hox c8* expression is reduced in the brain of mutants, consistent with *Asx1* being required both for activation and repression of *Hox* genes. We discuss the genetic and molecular definition of ETPs, and suggest that the function of *Asx1* depends on its cellular context.

Keywords

Hox; antero-posterior axis; epigenetic

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Introduction

To maintain the determined state, cells must pass on their gene expression patterns to their daughter cells in the absence of the transient signal or asymmetric localization that initiated the expression pattern. This process is epigenetic, because the change in gene expression is not accompanied by a change in DNA sequence (Cavalli, 2002). Two groups of genes are required to maintain gene expression patterns in higher eukaryotes. Polycomb group (PcG) proteins silence repressed genes, whereas trithorax group (trxG) proteins maintain activation (Brock and Fisher, 2005). These maintenance proteins (MPs) were originally discovered in *Drosophila*, and each of these has mammalian homologs (Brock and Fisher, 2005) (Schwartz and Pirrotta, 2008). The best known targets of MPs are the *Hox* genes that are required for axial patterning (Wellik, 2007). Genome-wide binding assays show that hundreds of other loci important for development, cell differentiation, signaling pathways, and some PcG genes themselves, are also targets (Bernstein et al., 2006; Boyer et al., 2006; Lee et al., 2006; Negre et al., 2006; Schwartz et al., 2006; Tolhuis et al., 2006). Mutations in MP genes have pleiotropic functions in axial patterning, stem cell differentiation, chromosomal position-effects, X-inactivation and imprinting, sex determination, chromosome segregation, RNA interference, organogenesis, hematopoiesis, and oncogenesis (Sparmann and van Lohuizen, 2006).

PcG and trxG proteins associate in complexes that alter histone modification, nucleosome structure, or nuclear structure (Muller and Verrijzer, 2009; Simon and Kingston, 2009; Brock and Fisher, 2005). PcG and trxG proteins may act by poising transcription (Mendenhall and Bernstein, 2008). Classically, animals bearing mutations of PcG genes exhibit posterior transformations in the antero-posterior (AP) axis that are enhanced in double mutants with other PcG genes. Similarly, trxG genes exhibit anterior transformations that are enhanced in animals mutant for two trxG genes.

However, this tidy division of maintenance proteins into two categories has been confounded by the discovery of mutations in genes encoding maintenance proteins that exhibit simultaneous anterior and posterior transformations and that enhance the homeotic transformations of both trxG and PcG mutations. These genes have been termed Enhancers of Trithorax and Polycomb (ETPs) (Brock and Lohuizen, 2001; Gildea et al., 2000). The phenotypes of ETPs imply that these genes function in maintenance of both activation and silencing of *Hox* genes. It has been proposed that the terms PcG and trxG be reserved for proteins that are members of complexes with demonstrated activity on chromatin, and that other proteins be classified either as proteins that recruit these complexes to their targets, or as trxG and PcG cofactors (Grimaud et al., 2006). Accordingly, ETPs could be subunits in both trxG and PcG complexes, recruit both trxG and PcG complexes, or be cofactors for both trxG and PcG complexes.

Additional sex combs (*Asx*) is an ETP in *Drosophila* that exhibits simultaneous anterior and posterior homeotic transformations (Breen and Duncan, 1986; Sinclair et al., 1992), and enhances homeotic phenotypes of both PcG (Sinclair et al., 1992) and trxG mutations (Milne et al., 1999). Mammals have three *Asx* homologs, termed *Additional sex combs-like 1* (*Asxl1*), *Asxl2*, and *Asxl3* (Fisher et al., 2003; Fisher et al., 2006; Katoh, 2003; Katoh, 2004). *Asx*-like proteins contain only two features conserved with *Asx*: an amino terminal region termed ASXH/ASXM containing two putative nuclear receptor co-regulator binding (NR box) motifs; and a carboxy-terminal PHD domain (Fisher et al., 2003); (Katoh, 2003). Additional sequence features are conserved within mammalian but not *Drosophila* *Asx* homologs, including three additional carboxy-terminal NR boxes (Fisher et al., 2006; Katoh, 2003; Katoh, 2004).

We recently generated a loss-of-function *Asxl1* mutant mouse model, *Asxl1^{tm1Bc}*, and used it to demonstrate that mouse *Asxl1* is required for normal adult hematopoiesis (Fisher et al.,

submitted). Here we employ the *Asxl1^{tm1Bc}* mouse mutant to determine if *Asxl1* demonstrates conserved ETP function. As predicted, mutant embryos exhibit simultaneous anterior and posterior homeotic transformations of the AP axis, consistent with our observations of *Hox* gene misregulation along the AP axis. We show that *Asxl1* is required for activation and repression of *Hox* genes. We discuss the definition of ETP in mammals, and propose that *Asx*-like protein function depends on cellular context.

Materials and Methods

Gene targeting and generation of *Asxl1^{tm1Bc}* mice

The null *Asxl1^{tm1Bc}* mutant allele was generated using a replacement targeting vector which inserted a PGK promoter-driven *neomycin* resistance cassette into exon 5 of *Asxl1* (Fisher et al., submitted), which is upstream of the conserved ASXH and PHD domains in *Asxl1* (Fisher et al., 2006). Genotyping was routinely performed on DNA from adult tail tips, liver from newborns, or yolk sac from embryos, by PCR as described (Fisher et al., submitted). All experiments with animals were performed in accordance with the regulations established by the Canadian Council on Animal Care, and all protocols were reviewed by the UBC Animal Care Committee.

Screening of *Cbx2^{tm1Ykf}* mice and generation of *Asxl1^{tm1Bc}; Cbx2^{tm1Ykf}* compound mutants

Cbx2^{tm1Ykf} (*M33^{Cterm}*) mice (Katoh-Fukui et al., 1998b) on a C57BL/6J background for at least N11 generations were crossed to mice heterozygous for *Asxl1* to generate double heterozygous mice. *Asxl1^{+/tm1Bc}; Cbx2^{+/tm1Ykf}* mice were interbred to generate other mutant allele combinations. Genotyping was routinely performed on DNA from adult tail tips, liver from newborns, or yolk sac from embryos, by triple primer PCR as follows. Primer M33A (5' GTAGCCAAGCCAGAGCTGAA 3') and antisense primer M33B (5' AGAGGCCTCTTTGGTGTGG 3') amplify a 200 bp fragment of the wild-type allele, and primers PGKPr (5' CCGCTTCCATTGCTCAGCGGT 3') and M33B amplify a 325 bp fragment of the *Cbx2^{tm1Ykf}* mutant allele (data not shown).

Skeletal preparations

Whole mount skeletons of newborns were prepared and stained with Alizarin Red and Alcian Blue 8GX as described (van der Lugt et al., 1994).

Statistical Analyses

Experimental genotype ratios were compared to expected ratios using the Chi-squared test. Comparison of means between groups was done using the Student's *t*-test, and significance was defined as *p* values <0.05.

In situ RNA hybridization

In situ hybridization to E10.5 mouse embryos was carried out as previously described (Belo et al., 1997). The *Hoxa4* riboprobe was obtained from Yoshihiro Takihara (Takihara et al., 1997). The *Hoxa7* riboprobe was reverse transcribed from clone 7514185 obtained from IMAGE consortium. The *Hoxc8* riboprobe has been described previously (Hanson et al., 1999).

Results

Phenotypic analysis of *Asxl1^{tm1Bc}* mutant mice

Mice heterozygous for *Asxl1^{tm1Bc}* exhibited Mendelian segregation ratios up to and just following birth. However, by postnatal day 21 (P21), only 28% of the expected number of

Asxl1^{tm1Bc/tm1Bc} mice was observed (Table 1). The majority of deaths occur within 1–3 days after birth. Most ill or dead *Asxl1^{tm1Bc/tm1Bc}* newborns and young pups had suckled and appear outwardly normal. The cause of death remains unclear as the mice are not anaemic (Fisher et al. submitted), and have normal lung, kidney, liver, and heart histology (data not shown). Adult *Asxl1^{tm1Bc/tm1Bc}* mice showed a significant reduction in body weight, splenomegaly, as well as a reduction in thymus and testis weight compared to wild-type mice (Fig. 1), but no change in weight of the liver, lungs, heart, or kidneys (data not shown). Four of 6 males and 3/3 female *Asxl1^{tm1Bc/tm1Bc}* mice tested were fertile. We occasionally noticed absence of one ovary in *Asxl1^{tm1Bc/tm1Bc}* mutant females (data not shown).

Homeotic transformations of the axial skeleton of *Asxl1^{tm1Bc}* mutants

We examined *Asxl1^{tm1Bc}* newborns (P1) for skeletal defects (Fig. 2, Table 2). The anterior arch is normally associated with only the C1 vertebra via a cartilaginous connection (Fig. 2A). In 13/14 of *Asxl1^{tm1Bc/tm1Bc}* mice the anterior arch was larger than in wild-type mice, and is additionally associated with the C2 vertebra via an ectopic cartilaginous process (black arrow in Fig. 2B), consistent with a C2 to C1 anterior homeotic transformation. In 2/14 of *Asxl1^{tm1Bc/tm1Bc}* mice the first cervical (C1) vertebrae was split laterally, similar to the break in ossification observed between the exoccipital and supraoccipital bones of the skull (Fig. 2B, grey arrow). This is interpreted as a C1 to occipital bone anterior transformation. An ectopic complete rib (2/14) or partial rib bud (3/14) on the C7 vertebra was seen in *Asxl1^{tm1Bc/tm1Bc}* (Fig. 2E), but not in wild-type mice (Fig. 4D), interpreted as a C7 to T1 posterior homeotic transformation. We observed small holes in the xiphoid process in 7/14 *Asxl1^{tm1Bc/tm1Bc}* mice, but not in wild type mice (data not shown). *Asxl1^{tm1Bc/tm1Bc}* mice (9/14) lacked the 13th thoracic (T13) rib on one or both sides, or had rib buds instead of complete ribs on one or both of the T13 vertebrae (Fig. 2G), indicating posterior transformation of thoracic T13 towards a lumbar (L1) vertebral identity. We did not observe sacral region transformations or defects in patterning of the appendicular skeleton in *Asxl1^{tm1Bc/tm1Bc}* mutants.

Asxl1^{Tm1Bc}; Cbx2^{tm1Ykf} compound mutant mice exhibit more severe phenotypes than single mutants

In *Drosophila*, ETP mutations enhance phenotypes of PcG mutations. Therefore we investigated the effects of a mutation in the PcG gene *Cbx2/M33* on *Asxl1^{tm1Bc}* phenotypes (Fig. 2). At day P21 the frequency of surviving *Asxl1^{+/+}; Cbx2^{+/+} tm1Ykf* mice (63%) agreed with that of *Cbx2^{+/+} tm1Ykf* (68%) mutants found previously (Katoh-Fukui et al., 1998b). Similarly, survivorship frequencies of *Asxl1^{+/+} tm1Bc; Cbx2^{+/+}* mice at P1 and P21 were consistent with the frequency of single *Asxl1^{+/+} tm1Bc* mutants (Table 1 and Table 3). Therefore background effects in the compound mutant were not significantly altering survivorship rates due to the *Asxl1^{tm1Bc}* mutation. However, by day P21, the genetic background does appear to affect viability of the *Cbx2 tm1Ykf/tm1Ykf* mutants, as we found this genotype at 6% frequency, versus 60% seen in the original study (Katoh-Fukui et al., 1998a).

Asxl1^{+/+} tm1Bc; Cbx2^{+/+} tm1Ykf trans-heterozygotes showed enhanced lethality compared to single mutants at P21 (Table 3). Strikingly, no *Asxl1^{tm1Bc/tm1Bc}; Cbx2 tm1Ykf/tm1Ykf* double homozygous mutant mice were seen at P21. Nor were double homozygous mutants found alive at P1 or E10.5–11.0, showing that the lethal phase occurs prior to E10.5–11.0. Consistent with this, four resorbed decidua were observed at E10.5–11.

We examined skeletal defects in *Asxl1^{tm1Bc}; Cbx2 tm1Ykf* compound mutant mice at P1 to detect enhancement of homeotic phenotypes (Fig. 2, and Table 2). In the *Asxl1^{+/+} tm1Bc; Cbx2^{+/+} tm1Ykf* trans-heterozygotes (n=11), 82% exhibited a C2 to C1 transformation (Fig. 4C), compared to 32% of *Asxl1^{+/+} tm1Bc* mutants (n=19) and 67% of *Cbx2^{+/+} tm1Ykf* mutants (n=6). The two *Asxl1^{tm1Bc/tm1Bc}; Cbx2^{+/+} tm1Ykf* mice analyzed both showed a C2 to C1 transformation.

As well, 64% of *Asxl1*^{+/tm1Bc}; *Cbx2*^{+/tm1Ykf} trans-heterozygotes exhibited sternal defects, including an abnormal xiphoid process and defective ossification in the 5th sternebra compared to only 33% of *Cbx2*^{+/tm1Ykf} mutants and 5% of *Asxl1*^{+/tm1Bc} mutants. The two *Asxl1*^{tm1Bc/tm1Bc}; *Cbx2*^{+/tm1Ykf} mice exhibited an offset joining of ribs to the sternum referred to as a 'crankshaft' sternum, and one of the mice had only 6 vertebrosteral ribs attached to the sternum on the left side instead of 7 as occurs in a wild type mouse (Fig. 2H, I). These two mice also had reduced or absent ossification centers (vertebral bodies) in some of the cervical vertebrae on the ventral side (not shown). A similar phenotype was observed in an *Asxl1*^{+/tm1Bc}; *Cbx2*^{+/tm1Ykf} mouse found dead shortly after birth. In addition, this mouse also had a split scapula, and a fusion between the atlas and the exoccipital bones of the skull (Fig. 2C). Similar sternal phenotypes have been observed in *Cbx2*^{tm1Cim/tm1Cim} (van der Lugt et al., 1996) and *Cbx2*^{tm1Ykf/tm1Ykf} mice (Katoh-Fukui et al., 1998b) but not in *Cbx2* or *Asxl1* heterozygous mutant mice, indicating enhancement of these phenotypes in the *Asxl1*; *Cbx2* mutant trans-heterozygotes. The penetrance of T13 to L1 transformations was increased in the *Asxl1*^{+/tm1Bc}; *Cbx2*^{+/tm1Ykf} trans-heterozygotes (45%) compared to 26% in *Asxl1*^{+/tm1Bc} and 17% in *Cbx2*^{+/tm1Ykf} mutants. Overall, our results show an enhancement of bidirectional skeletal homeotic transformations in compound *Asxl1*; *Cbx2* mutants compared to single *Asxl1* and *Cbx2* mutants.

Hox gene expression in *Asxl1* mutant mice

The skeletal transformations observed in *Asxl1*^{tm1Bc} mutants are consistent with mis-expression of *Hox* genes. Therefore we analyzed embryos for changes in the pattern of mRNA expression of *Hoxa4*, *Hoxa7* and *Hoxc8* in mutant and wild-type E10.5 embryos by whole mount in situ hybridization (Fig. 3). These *Hox* genes show ectopic expression in *rae28*, *bmi-1*, *mel-18* or *M33* mutants, and reduced expression in *Mll* mutants, and thus provide a basis for comparison to the results obtained in *Asxl1*^{tm1Bc} mutants. In wild-type embryos, the anterior boundary of *Hoxa4* ectodermal expression is detected at the upper cervical spinal cord and spinal ganglia, and expression levels decrease caudally until it is undetectable at the thoracic level. In mesoderm, *Hoxa4* is expressed anterior to prevertebra 1, and is reduced posterior to prevertebra 6 (Toth et al., 1987). In 5/7 homozygous *Asxl1*^{tm1Bc} embryos, the anterior boundaries of both ectodermal and mesodermal expression are shifted anteriorly by 1 or 2 somites relative to wild-type (Fig. 3; blue and red arrows respectively). Two of these 5 mutants also show *Hoxa4* expression in caudal mesoderm where *Hoxa4* expression is never detected in wild type.

In wild-type mice, ectodermal *Hoxa7* expression begins at the C5 somite and ends at the 4th sacral (S4) somite, whereas mesodermal expression begins at T3 and ends at T13 (Dressler and Gruss, 1989), (Puschel et al., 1991). Anterior boundaries of *Hoxa7* in ectoderm and mesoderm are shifted rostrally in 8/8 homozygous *Asxl1*^{tm1Bc} mutants by 1 somite to C4 and T2 respectively (Fig. 3). These results show that *Hoxa4* and *Hoxa7* are derepressed in *Asxl1*^{tm1Bc} mutants.

The effect of *Asxl1*^{tm1Bc} mutation on *Hoxc8* expression in embryos is more complex. *Hoxc8* expression is shifted anteriorly in ectoderm and mesoderm of 8/8 homozygous *Asxl1*^{tm1Bc} mutants by 1 somite (Fig. 3). The overall intensity of staining in the ectoderm and mesoderm of mutant embryos is much greater than in wild-type embryos, and expression of *Hoxc8* can be detected more caudally than in wild type, consistent with a role for *Asxl1* in repressing *Hoxc8* in ectoderm and mesoderm. However, in the developing brain, *Hoxc8* expression is consistently reduced in mutant embryos compared to wild type embryos where it is strongly expressed (Kwon et al., 2005), suggesting that *Asxl1* is required for activation of *Hoxc8* in the brain. The mutant embryos cannot simultaneously be overstained for expression in somites

and spinal cord, and understained in the brain, so the differences between the *Asxl1^{tm1Bc}* mutant and wild-type embryos do not reflect technical problems with staining.

Discussion

The homeotic transformation phenotypes of the *Asxl1^{tm1Bc}* mutants are mild relative to that of most other PcG mutants in mice. The hybrid genetic background decreases the severity of the *Asxl1^{tm1Bc}* phenotypes reported here, as further backcrossing to C57BL/6J has resulted in complete embryonic lethality of homozygous mutants (C.F, unpublished). Similar effects have been seen in mice mutant for *Mel18/Pcgf2* upon further backcrossing to a consistent background (Suzuki et al., 2002). Murine *Asxl2^{Gt(AQ0356)}* gene-trap mutants on a mixed genetic background also exhibit incompletely penetrant perinatal lethality (Baskind et al., 2009). As *Asxl1* and *Asxl2* have similar, broad expression patterns in mouse embryos and stem cells (Fisher et al., 2006), redundancy of *Asxl2* may mask early functions of *Asxl1*. Embryonic stem cell expression of other MPs correlates with early developmental function (Ayton et al., 2001; Donohoe et al., 1999; O'Carroll et al., 2001; Voncken et al., 2003). The generation of compound *Asxl1*; *Asxl2* mutant mice will allow testing of potential redundancy, and of early developmental roles for *Asxl* genes.

Anterior transformations are usually associated with loss of function *Hox* mutations, and thus the C2 to C1 and C1 to occipital bone transformations observed in *Asxl1^{tm1Bc}* mice suggest that *Asxl1* is required for *Hox* activation. Anterior homeotic transformations in the C1 and C2 cervical vertebrae are also seen in *Hoxb4*, *Hoxd3*, and *Hoxd4* mutant mice (Condie and Capecchi, 1994; Horan et al., 1995). However, in some cases the complexity of the *Hox* code means that loss of function *Hox* mutations can cause posterior transformations, as in the case of C7 to T1 thoracic vertebral transformations observed in *Hoxa4* (Horan et al., 1994), *Hoxa5* (Jeannotte et al., 1993), or *Hoxa6* knockout mice (Kostic and Capecchi, 1994). Therefore the C7 to T1 transformation in *Asxl1^{tm1Bc}* mutants could also be consistent with an activating role for *Asxl1*.

In most cases, posterior homeotic transformations are associated with derepression of *Hox* expression (Pollock et al., 1995). This is the case for the transformations of C7 to T1, and T13 to L1 vertebrae observed in *Asxl1^{tm1Bc}* mutant embryos. The observations that *Hoxa4*, *Hoxa7* and *Hoxc8* are derepressed, and that *Asxl1^{tm1Bc}* mutations enhance the phenotypes of *Cbx2* mutations are also consistent with a role for *Asxl1* in silencing of *Hox* genes. Together, the anterior and posterior transformations of *Asxl1^{tm1Bc}* mutants suggest that *Asxl1* is an ETP. Consistent with this conclusion, we observed a role for *Asxl1* in both silencing and activation of *Hoxc8*.

It may be that ETP phenotypes of *Asxl1* are indirect consequences of *Asxl1*-mediated gene expression. The homeotic transformations we observe could arise because *Asxl1* is a co-regulator of RAR in the cervical vertebrae. Mice in which retinoid receptor function is impaired by gene targeting (Mark et al., 2006) or by dominant negative mutations (Yamaguchi et al., 1998), exhibit anterior transformations of cervical vertebrae. This role for *Asxl1* is also consistent with the enhancement of anterior phenotypes observed in *Asxl1^{tm1Bc}*; *Cbx2^{tm1Ykf}* double mutants because *Cbx2* antagonizes the RA pathway and functions in establishing the early spatiotemporal sequence of activation of a subset of *Hox* genes (Bel-Vialar et al., 2000). However, posterior derepression of *Hox* genes, and posterior transformations are inconsistent with a direct consequence of *Asxl1* acting through the RAR, because defects in RAR signaling cause anterior transformations (Daftary and Taylor, 2006).

Indirect phenotypes could also arise because *Asx* and some *Drosophila* PcG genes positively regulate each other's expression. *Cbx2* is a known member of the mammalian Polycomb group

Repressing Complex 1 family (Simon and Kingston, 2009) yet mutants exhibit both anterior and posterior transformations (Core et al., 1997; Katoh-Fukui et al., 1998b). In addition, the Sex combs extra homologs Ring1/Ring1a and Rnf2/Ring1b show opposing loss-of-function phenotypes in mutant mice despite both gene products being subunits of PRC1. Homozygous *Ring1^{tm1Mvi}* mutants exhibit anterior transformations (del Mar Lorente et al., 2000), whereas *Rnf2^{tm1Hko}(Ring1b^{red})* hypomorphic mutants exhibit posterior transformations (Suzuki et al., 2002). These observations suggest the need for caution when inferring molecular function from phenotypic analysis (Grimaud et al., 2006).

An alternative explanation for the ETP phenotypes of *Asx11* mutations is that *Asx11* function is context-dependent. In this model, *Asx11* is required in the trunk to silence *Hoxa4*, *a7*, and *c8*, but is required in the brain to activate *Hoxc8*. In *Drosophila*, *Asx* is unusual among PcG genes because *Asx* mutants exhibit tissue-specific effects (Soto et al., 1995) even though expression is ubiquitous (Sinclair et al., 1998), which would be consistent with context-dependent function. In support of this interpretation, in human HeLa cells, mouse *Asx11* tethered to GAL4 binding sites activates a heterologous thymidine kinase promoter driving a luciferase reporter, and also increases RAR-dependent-activation of the reporter in transient transfection assays; however, in other mammalian cell types, *Asx11* acts as a RAR corepressor (Cho et al., 2006). In addition, *Asx11* is a member of a repressive complex containing Histone H1.2 (Kim et al., 2008).

Interestingly, *Asx12^{Gt(AQ0356)}* mutants exhibit anterior transformations of thoracic vertebrae at low penetrance, and highly penetrant posterior transformations of sacral vertebrae (Baskind et al., 2009). These mutant phenotypes are consistent with expectations for an ETP although as they are based upon a single gene-trap mutant model they should be confirmed in an independent *Asx12* null mutant line. Since sacral region transformations were never seen in *Asx11^{tm1Bc}* mice, this suggests that a functional division has occurred whereby *Asx11* predominantly regulates more anterior cluster Hox genes, while *Asx12* function is more focused on the posterior cluster Hox genes, albeit with substantial overlap in regulation of mid-cluster Hox genes. In addition, there is a pattern in both *Asx11* and *Asx12* mutants whereby the more rostrally located axial vertebrae undergo anterior transformation, while the more caudally located vertebrae demonstrate posterior transformations. This could imply a mechanistic role of *Asx*-like proteins in the activation of more rostral Hox genes, but in repression of more caudally located Hox genes, within the functional domain of each respective *Asx*-like family member. We propose that *Asx11* function is determined by its interacting partners or chromatin environment to bring about changes in gene regulation. Identifying the target genes and interaction partners of *Asx11* and other *Asx*-like proteins, and how they bring about changes in gene regulation remains an important challenge for the future.

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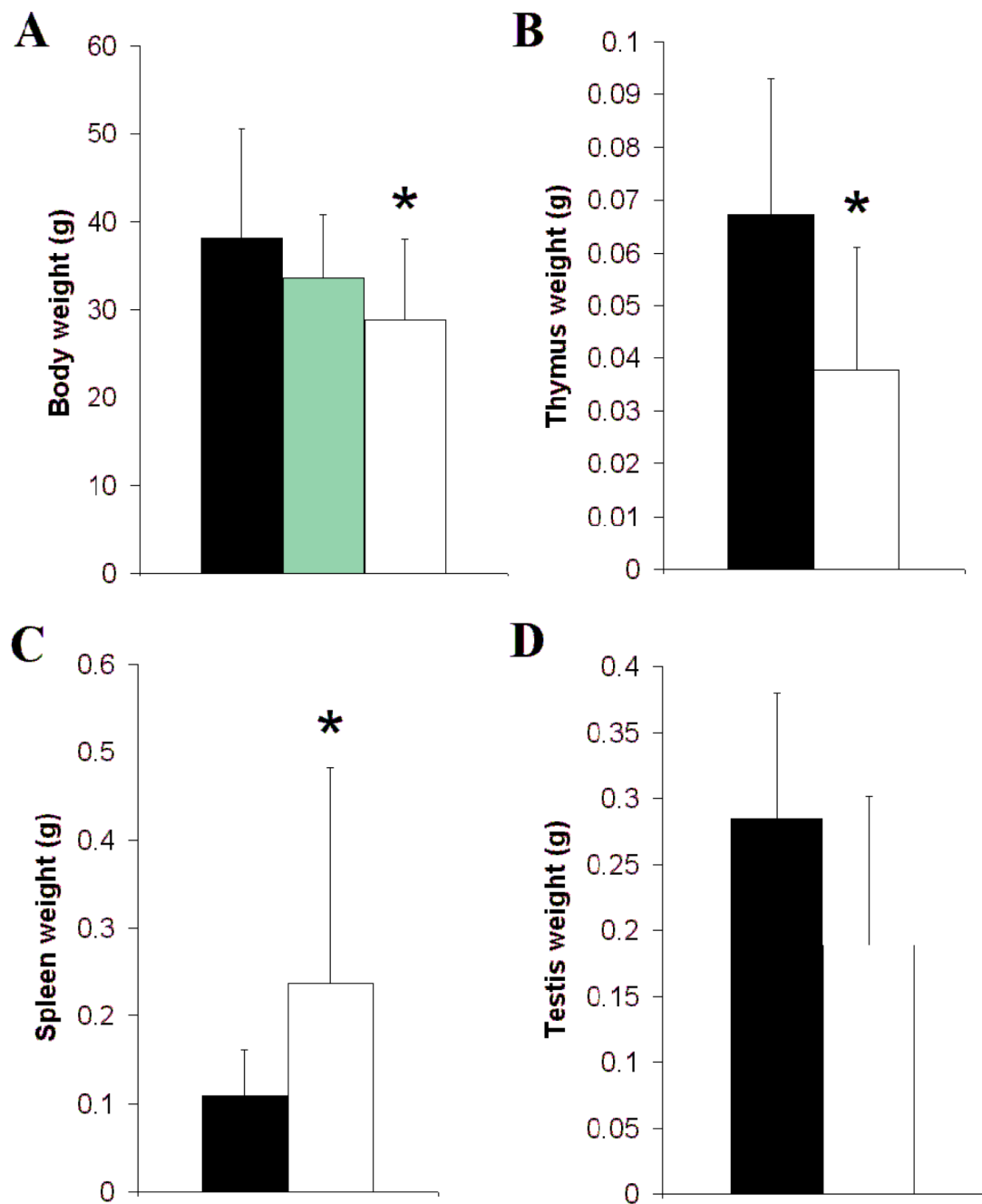


Fig. 1. *Asxl1^{tm1Bc}* mutant adult mice exhibit reduced body and testis weight, hypothyroidism, and splenomegaly. (A) Body weight is significantly reduced in *Asxl1^{tm1Bc}* mutant (mut) mice (n=12 wt, 5 +/-mut, 12 mut/mut). (B) *Asxl1^{tm1Bc/tm1Bc}* mice are hypothyroid (n=11). (C) *Asxl1^{tm1Bc/tm1Bc}* mice exhibit splenomegaly (n=12). (D) *Asxl1^{tm1Bc/tm1Bc}* mice have reduced testis weight (n=5 +/-, 6 mut/mut; p=0.08). *, p<0.05; average values are shown with error bars indicating one SD. Black bars indicate samples from wild type; grey bars heterozygotes; white bars *Asxl1^{tm1Bc/tm1Bc}* mutants.

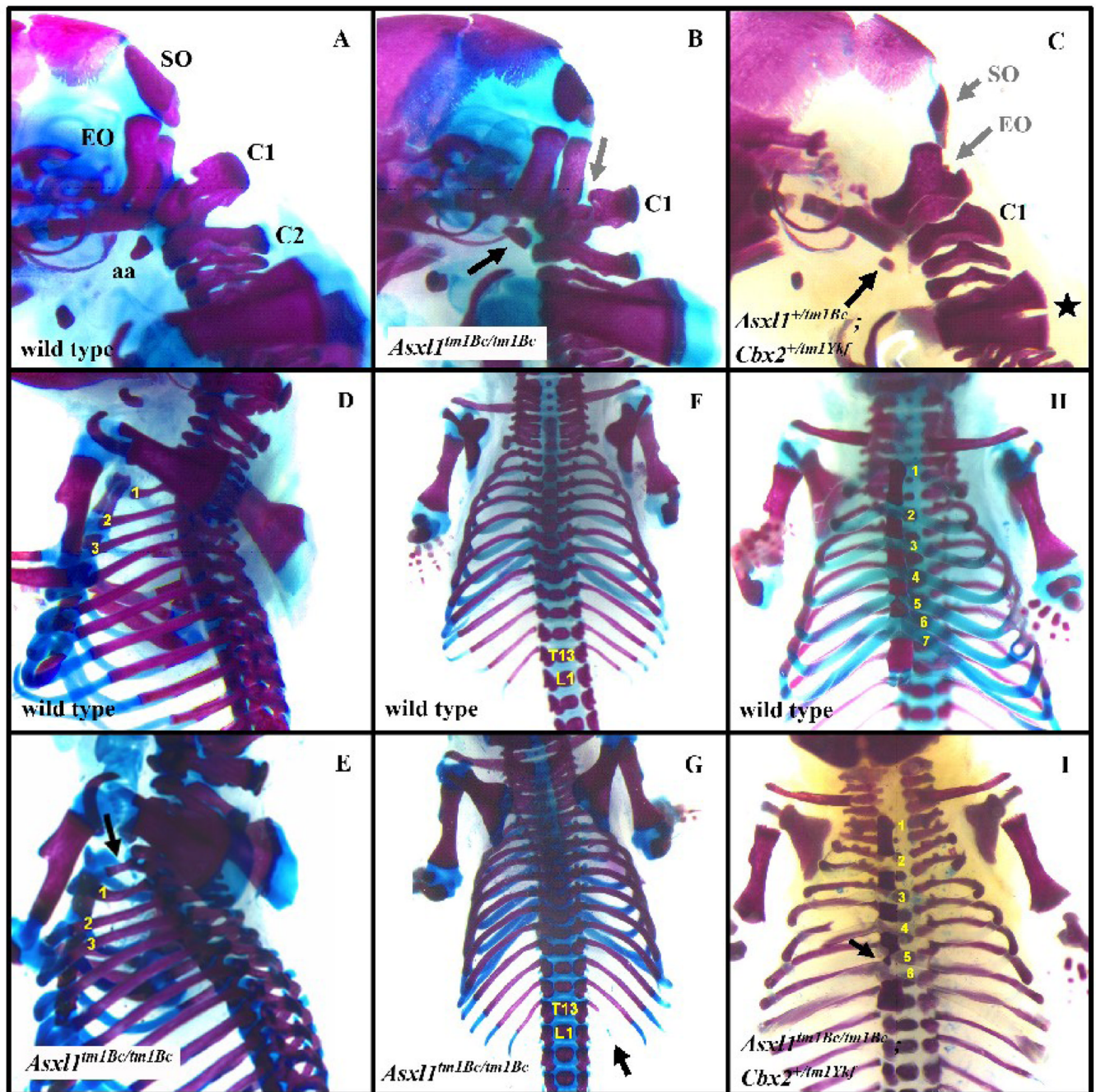


Fig. 2.

Alterations of the axial skeleton of newborn *Asx1^{tm1Bc}* mutant and *Asx1^{tm1Bc}; Cbx2^{tm1Ykf}* compound mutant mice. Bone and cartilage were stained with alizarin red and alcian blue respectively. Lateral views of the cervical regions and scapulae of cleared skeletons (A–C). (A) Wildtype mouse with cervical 1 (C1), cervical 2 (C2), anterior arch (aa), supraoccipital (SO) and exoccipital (EO) bones indicated. Notice the association of the aa with C1 via cartilage, the intact C1, and the separation between the SO and EO. (B) *Asx1^{tm1Bc/tm1Bc}* mouse. The C2 vertebra is transformed towards the C1 vertebra shown by an association of the now larger aa with C2 (black arrow) and C1. The C1 vertebra is anteriorly transformed toward the SO and EO bones of the skull indicated by a split within the C1 vertebra (grey arrow). (C)

Asxl1^{+/tm1Bc}; *Cbx2*^{+/tm1Ykf} compound mutant. The C2 vertebra is anteriorly transformed towards the C1 vertebra, while the aa is reduced in size (black arrow). More severe anterior transformations are indicated by fusion of the C1 vertebra to the EO (long grey arrow). The SO is reduced (short grey arrow); the scapula (star) exhibits defective ossification. Lateral views of the thoracic regions of cleared skeletons, where the first 3 thoracic (T) vertebrae are numbered (D–E). (D) Wild-type mouse. (E) *Asxl1*^{tm1Bc/tm1Bc} mouse with a C7 to T1 posterior transformation is indicated by an ectopic rib on the C7 vertebra, shown with a black arrow. Dorsal views of the thoracic and upper lumbar (L) regions of cleared skeletons (F–G). (F) Wildtype mouse, showing two ribs on T13, and absence of ribs on L1. (G) *Asxl1*^{tm1Bc/tm1Bc} mouse exhibiting size reduction of one T13 rib, noted with a black arrow. The cartilage associated with the T13 rib is still present. Ventral views of the thoracic region of cleared skeletons (H–I). (H) Wildtype mouse showing 7 normal vertebrosteral ribs numbered in yellow. (I) *Asxl1*^{tm1Bc/tm1Bc}; *Cbx2*^{+/tm1Ykf} mutant mouse with defects in the sternum, notably reduced ossification of the 5th sternebra (black arrow), and the presence of only 6 vertebrosteral ribs attached to the sternum on the left side.

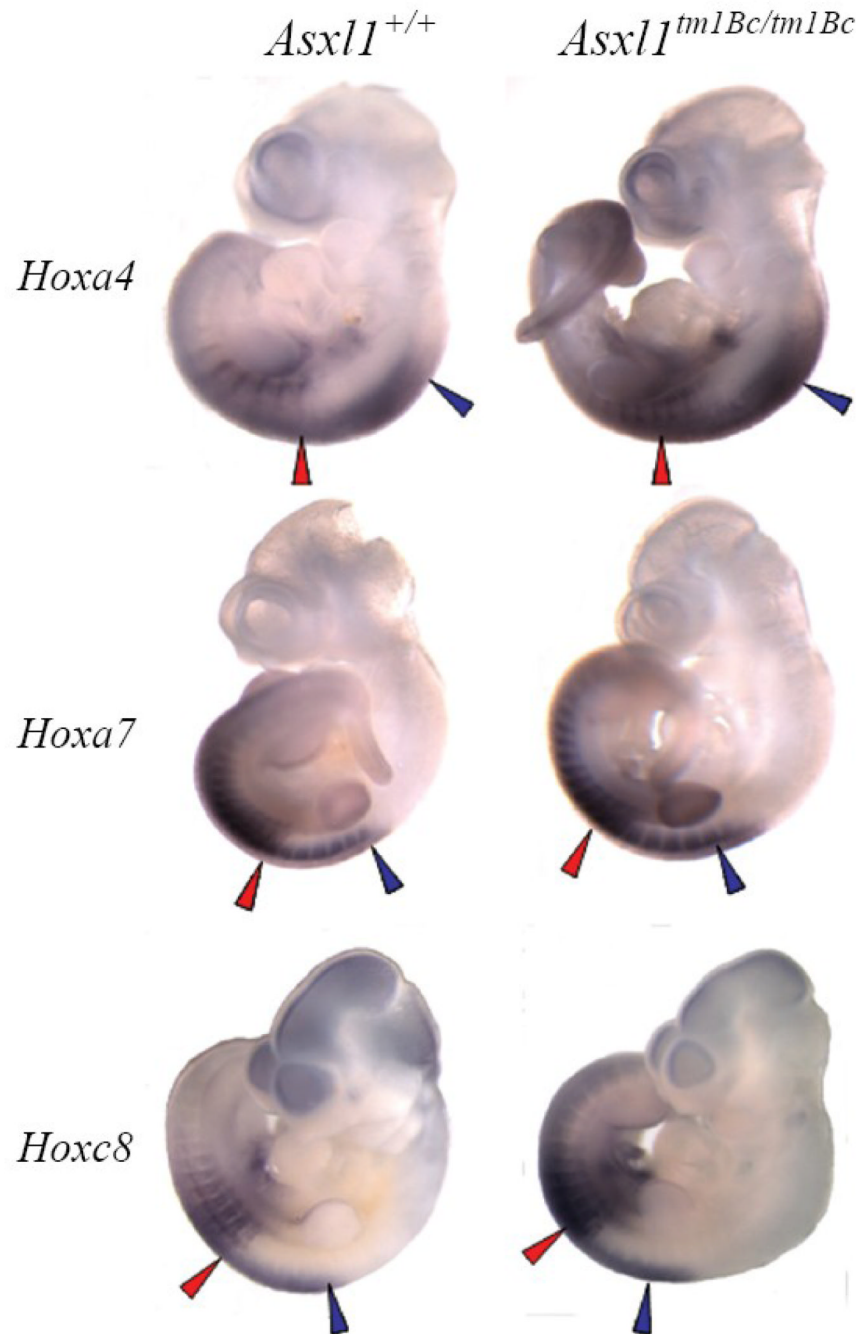


Fig. 3. *Asxl1*^{tm1Bc/tm1Bc} embryos exhibit altered *Hox* expression. In situ hybridization of *Hoxa4*, *Hoxa7*, and *Hoxc8* riboprobes to wildtype and *Asxl1*^{tm1Bc/tm1Bc} mutant E10.5 embryos. Anterior boundaries of ectoderm (blue) and mesoderm (red) expression are shown by coloured arrowheads.

Table 1

Numbers of offspring at different ages from heterozygous *Asx1* intercrosses.

<i>Asx1</i> genotype	E10.5–11.5 n=50	P1 n=119	P21 n=463
+/+	15	31	213
+/tm1Bc	22	54	370
tm1Bc/tm1Bc	12	26	60

The observed frequencies are significantly different ($p < 0.05$, Chi-square test) from expected only at 21 days after birth (P21).

Table 2
Percentages of skeletal alterations observed in newborn *Asx11* and *Asx11*; *Cbx2* compound mutants.

Phenotype	Genotype					
	+/+ n=11	Asx11/+ n=19	Asx11/Asx11 n=14	+;Cbx2/+ n=6	Asx11/+;Cbx2/+ n=11	Asx11/Asx11; Cbx2/+ n=2
C1 to exoccipital	0	0	14	0	9	0
C2 to C1	9	32	93	67	82	100
C7 to T1	0	0	14	17	0	0
Abnormal scapula	0	0	0	0	9	0
T7 to T8	0	0	0	0	9	50
Offset ribs	0	0	0	0	27	100
Abnormal xiphoid process	9	5	50	33	64	100
T13 to L1	9	26	64	17	45	100

Penetrance of the skeletal abnormalities expressed as a percentage of the embryos analyzed for each genotype. Abbreviations and phenotypes are described in the text. The expressivity of the C1 to occipital transformation is much more severe in *Asx11*; *Cbx2* double mutants than in *Asx11* homozygotes (see Fig. 4C).

Table 3

Numbers of offspring of intercrosses between compound heterozygous *Asxl1* and *Cbx2* mutant mice by genotype and age.

Genotype	10.5–11.5 n=39	P1 n=33	P21 n=227
<i>Asxl1</i> ^{+/+} ; <i>Cbx2</i> ^{+/+}	2	2	36
<i>Asxl1</i> ^{+/tm1Bc} ; <i>Cbx2</i> ^{+/+}	5	6	63
<i>Asxl1</i> ^{+/+} ; <i>Cbx2</i> ^{+/tm1Ykf}	1	12 (1)	46
<i>Asxl1</i> ^{+/tm1Bc} ; <i>Cbx2</i> ^{+/tm1Ykf}	14	9 (1)	67
<i>Asxl1</i> ^{tm1Bc/tm1Bc} ; <i>Cbx2</i> ^{+/+}	1	0	3
<i>Asxl1</i> ^{+/+} ; <i>Cbx2</i> ^{tm1Ykf/tm1Ykf}	2	1 (1)	2
<i>Asxl1</i> ^{tm1Bc/tm1Bc} ; <i>Cbx2</i> ^{+/tm1Ykf}	4	3 (1)	7
<i>Asxl1</i> ^{+/tm1Bc} ; <i>Cbx2</i> ^{tm1Ykf/tm1Ykf}	4	0	1
<i>Asxl1</i> ^{+/tm1Bc} ; <i>Cbx2</i> ^{tm1Ykf/tm1Ykf}	4	0	1

The number of newborn mice within each genotype group found dead are noted in round brackets. The observed frequencies are significantly different from expected ($p < 0.05$, Chi square test) only for 21 days after birth (P21). Numbers for the other age categories are too low to analyze using the Chi square test.